2(a)	**M1** Al_2O_3 is ionic **M2** P_4O_{10} is a (simple) molecule **M3** ionic bonds are stronger than intermolecular forces	**3**

2(c)(i)	increases rate by providing a different mechanism with a lower activation energy	**1**

Examiner report missing
9701_w25_qp_24 Question 2

3(a)(i)	Al_2O_3 / aluminium oxide	**1**
3(b)(i)	nickel / platinum	**1**
3(b)(ii)	minimum energy required for a collision to be effective	**1**
3(b)(iii)	**M1** ● area under the curve with energy $\geqslant E_A$ using a catalyst is larger so ● greater proportion of particles with sufficient energy to react **M2** rate faster because higher frequency of successful collisions	**2**

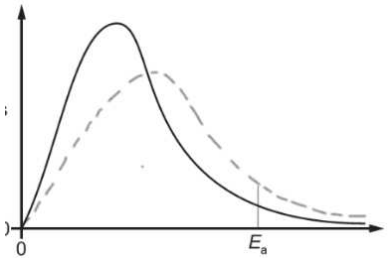
Examiner report missing
9701_s25_qp_23 Question 3

4(a)(i)	Check final graph for range of acceptable answers ● ±(1–0.5) / 100 = ±0.005 ● 0 ● mol dm^{-3} s^{-1} for both Any 2 points = 1 mark, all 3 points = 2 marks	**2**
4(a)(iii)	the frequency of effective collisions decreases as concentration of HC*l*(aq) and / or $Na_2S_2O_3$ decreases	**1**
4(a)(v)	It reacts with water (to produce H_2SO_3)	**1**

Examiner report missing
9701_s25_qp_23 Question 4

3(a)(i)	two (or more) reactants / molecules react to make (only) one product	**1**
3(b)(i)	H_2 / hydrogen **AND** some X / alkene left(over) / remains (at the end of this reaction)	**1**
3(b)(ii)	*link the significance of gradient values to rate **AND** describe either the decrease in rate or the decrease in gradient* as the gradient decreases the rate decreases **OR** rate decreases until the reaction stops when gradient = 0	**1**
3(d)(ii)	**M1** it / (area) **B** larger **AND** rate (reaction / hydrogenation) increases **OR** (area) **B** identified by shading or labelling on Fig 3.3 **AND** rate (reaction / hydrogenation) increases in prose **M2** increase in the frequency of effective / successful collisions **OR** increase in rate of effective / successful collisions	**2**

Examiner report missing
9701_s25_qp_22 Question 3

3(a)(i)	 *Two correct for one mark, all four for two marks:* • start at origin • peak sits below the original • peak sits to the right-hand side of the original • new line crosses the original once only **AND** to the right of the original peak	**2**
3(a)(ii)	**M1** greater proportion of (H_2) molecules have energy greater than or equal to the activation energy / ≥ E_A **M2** increase in frequency of successful collisions (between H_2 and C_6H_{12})	**2**
3(a)(iii)	catalyst	**1**
3(a)(iv)	annotation to show new E_A (nickel / catalyst) further to the left than original E_A	**1**

Examiner report missing
9701_s25_qp_21 Question 3

2(a)	58% / no change / stays the same **AND** catalysts don't affect yield / iron is a catalyst	**1**
2(b)(i)	less yield (in **C** than **A**) **AND** (as) exothermic (forward reaction)	**1**
2(b)(ii)	(increase in temperature / energy from **A** to **C** (so)) **M1** more collisions with particles that have $E > E_a$ **AND** increase frequency of effective collisions **M2** (initial) rate (of the forward reaction) increases	**2**

Question 2

Although **(d)** focused on inorganic chemistry, the majority of this question was based on equilibria and rates. Better performing candidates were able to discern the differing aspects here: rates and collision theory formed one notion; equilibria, yield and Le Chatelier's principle the other.

(a) This was generally well answered, with many candidates identifying the null effect of a catalyst on equilibrium position. The best answers stated this explicitly.

(b)(i) This was well answered by candidates, who could link lower yield with the exothermic reaction. Precision of wording was key here.

(ii) Many candidates attempted to answer this rates question in terms of equilibrium yield and were unable to gain full credit as a result. If an item pairs rate and yield within a question, candidates are advised to read the stem of the question carefully and to give relevant answers.

(c)(i) Many candidates were unable sufficiently to distinguish between K_c and K_p. Square brackets must not be used in K_p expressions to avoid confusion with concentration variables. Partial pressure variables can be given as p_X, P_X, $p(X)$, $P(X)$ or $pp(X)$.

(ii) Many numerical answers were correct, though common errors included not adjusting the equilibrium concentrations of N_2 and H_2, or not accounting for **2**NH_3 or **3**H_2 in the stoichiometric equation.

(iii) This was generally well answered by candidates, although some errors of powers of ten were seen in a minority of responses.

(d)(i) Candidates' knowledge of this nitrogen chemistry seemed largely secure, though correct quotation of three equations (two showing NO_2 catalysis and one of the conversion of SO_3 to acid rain) were needed to gain full credit, alongside a statement of the role of NO_x.

(ii) There were many good partial answers to this question, with most referring to PAN but with less precision on how this is produced.

3(a)	strong triple bond / high triple bond enthalpy	**1**
	non-polar (molecule)	**1**

3(c)(i)	0 for both ΔH_f of N_2 and H_2	**1**
	−46 for ΔH_f of NH_3	**1**

3(c)(ii)	increases rate by providing a different mechanism with a lower activation energy	**1**

3(c)(iii)	rate is lowered	**1**
	lower frequency of successful collisions / number of successful collisions per unit time	**1**

Question 3

This question tested candidates' abilities to apply their knowledge in slightly unfamiliar contexts. In-depth knowledge and recall of nitrogen chemistry was crucial, as was the ability to draw clear, labelled, unambiguous diagrams.

(a) Different types of answer were needed to gain full credit. A strong triple bond and high E_a are in effect making the same point.

(b) (i) There were many correct answers here, with balancing being the main difficulty.

(ii) Very few answers correctly referenced both acidic species, HNO_3 and HNO_2. The latter was rarely seen.

(iii) Candidates deployed a full range of answers referring to environmental issues: specifically, this was looking at photochemical smog.

(c) (i) Candidates found it challenging to apply the definition of standard enthalpy change of formation; attempting to conceive of two numbers that would sum to –46 (or –92), or that combined with the ΔH_f would sum to zero.

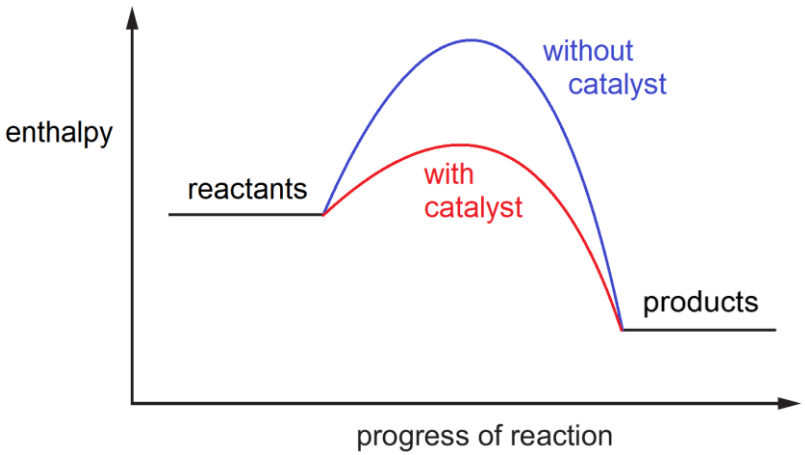
(ii) Many candidates were able to give good definitions here.

(iii) Candidates did not perform well on this question. Despite '**rate**' being in bold in the question, the large majority of responses attempted to answer in terms of yield and equilibrium position. Nevertheless, some credit was available for correct and non-contradicted information about a decrease in rate. Where collision theory is referenced, candidates are reminded of the need to discuss the *frequency* (i.e., number *per unit time*) of collisions, rather than simply the number.

(d) (i) Many diagrams showed a triple bond between the nitrogen atoms or miscounted the number of non-bonding electrons.

(ii) Candidates did not show a good understanding of hybridisation as responses divided reasonably equally between those who offered sp, sp^2 and sp^3 as answers.

(iii) Some good answers were seen here, but many answers could not be awarded full credit. *Sideways* overlap of p orbitals was needed in the response, or a clear and labelled diagram showing this as part of an unambiguous representation of a pi bond.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**
3(b)	 uncatalysed hump	**1**
	catalysed hump lower than uncatalysed	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

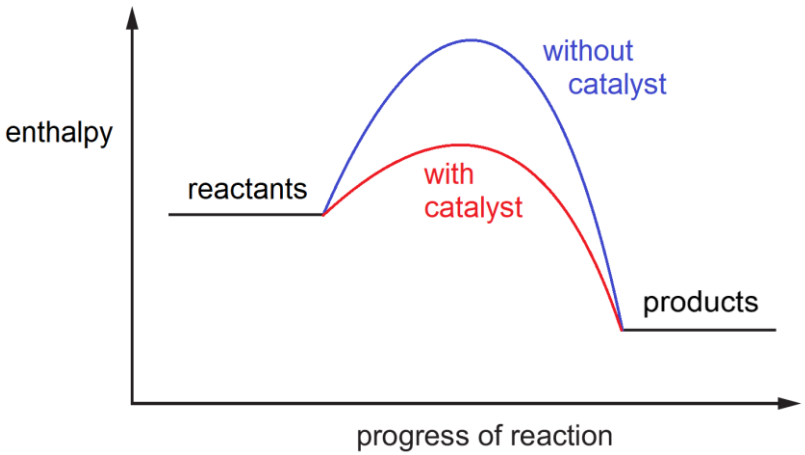
Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

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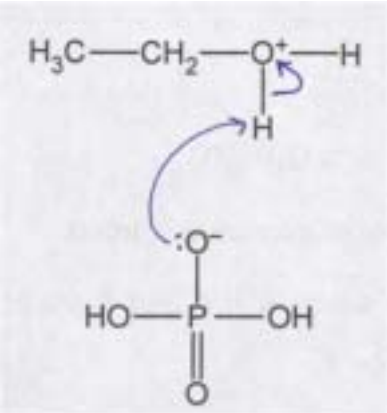
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(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

4(a)	Method 1 • identification of number and type of bonds when all are broken 2O-H + 4C-H + C=C **OR** (2 × 460) + (4 × 410) + 610 • Identification of number and type of bonds when all are made O-H + 5C-H + C-C + C-O **OR** (460 + (5 × 410) + 350 + 360) • calculation of energy to break bonds – energy to make bonds 3170 – 3220 = – 50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks. **OR** Method 2 • identification of number and type of relevant bonds broken (including the new C-H only) 2O-H + C=C **OR** (2 × 460) + 610 • identification of number and type of relevant bonds made (including the new C-H only) O-H + C-H + C-C + C-O **OR** (460 + 410 + 350 + 360) • calculation using energy to break bonds – energy to make bonds 1530 – 1580 = –50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks.	**2**
4(c)(i)	acid **AND** it donates a proton / H^+	**1**
4(c)(iii)	 **M1** arrow from bond of one of the O-H to O of the cation **M2** arrow from lone pair on O of the anion to between H of H-O	**2**

4(c)(iv)	**M1** a catalyst increases rate of reaction **M2** by producing an alternative reaction pathway with lower activation energy	**2**

4(c)(v)	it is regenerated **OR** it is reformed	**1**

Question 4

(a) Stronger candidates showed clear logical working and selection of appropriate data. Common errors included mis-reading the table, using an inappropriate value for a correctly identified bond, or ignoring the C-O bond during the reaction.

(b) Reference to the effect of increasing pressure on the amount of ethanol produced at equilibrium was not always included. Many explanations were given in terms of the change producing a shift to the position of equilibrium to the right; the significance of producing a smaller amount of gas to explain this change was mentioned less frequently.

(c) (i) Many candidates were able to deduce the behaviour of H_3PO_4 as an acid or an electrophile. Some explanations were not specific to step 1 in the reaction.

(ii) Some candidates were able to identify the correct species in step 2. Explanations were often vague, for example, 'seeking electrons' or 'being attacked by a pair of electrons' rather than 'accepting a lone pair'.

(iii) A small proportion of candidates applied knowledge of the drawing of mechanism arrows to represent electron movement when bonds are made and broken in this unfamiliar context. A common error showed arrows pointing in the opposite direction to the way an electron pair moves when a bond is made or broken. Sometimes an arrow did not start from the bond being broken or from the lone pair of electrons on an atom that is used to make a specific bond.

(iv) It was generally well known how a catalyst affects a reaction. Some ambiguity was seen in answers that described the effect of catalysts as increasing the number of collisions between particles rather than increasing the frequency of successful collisions.

(v) Stronger candidates clearly described the regeneration of the catalyst. Some candidates were vague and imprecise, for example 'the catalyst did not change in the reaction'.

(vi) Some candidates showed an excellent understanding of the consequence of more alkyl groups present on the C^+ of the intermediate in terms of a larger inductive effect. Some candidates incorrectly referred to the inductive effect of propan-2-ol rather than the carbocation intermediate which was responsible for the production of propan-2-ol.

(d) The sideways overlap of two p orbitals to form a pi bond between the two carbon atoms was well known. Descriptions of the formation of the sigma bond by direct overlap of sp^2 hybrid orbitals were less common.

5(a)(i)	$C_8H_{18} + 12½\ O_2 \rightarrow 8CO_2 + 9H_2O$ **M1** correct formula of M C_8H_{18} **M2** balanced equation showing complete combustion of M	**2**

5(b)(i)	**M1** ● high temperature ● high pressure ● N_2 Any two correct for **M1** **M2** $N_2 + O_2 \rightarrow 2NO$ **OR** $N_2 + 2O_2 \rightarrow 2NO_2$ **OR** $N_2 + xO_2 \rightarrow 2NO_x$	**2**

5(b)(ii)	**M1** NO_2 is the catalyst (for $SO_2 \rightarrow SO_3$) **OR** catalytic oxidation (of SO_2) by NO_2 **M2** $NO_2 + SO_2 \rightarrow NO + SO_3$ **AND** $2NO + O_2 \rightarrow 2NO_2$ **M3** SO_3 reacts with rain **OR** moisture/water (in atmosphere) to form sulfuric acid / H_2SO_4 **OR** $SO_3 + H_2O \rightarrow H_2SO_4$	**3**

Question 5

(a) (i) This was well answered by most candidates, although balancing the equation proved challenging for some.

(ii) Most candidates correctly mentioned the production of CO instead of CO_2. A very small number correctly referred to C and unburnt hydrocarbons.

(b) (i) A large proportion of candidates incorrectly interpreted this as a question about catalytic converters. Many answers used NO instead of N_2.

(ii) A small number of candidates correctly identified the catalytic role of NO_x and gave all the relevant equations. Many gave the equation for the formation of H_2SO_4, and a significant number gave equations and answers related to the formation of HNO_3 for acid rain.

(iii) This was well answered by most. Some candidates gave the identity of a gas instead of the type of air pollution.

(c) (i) Candidates often omitted dilute/(aq) with a named acid here.

(ii) A few candidates recognised that all three ester groups will undergo hydrolysis to form propanetriol. Many answers gave fragments where only one ester group had undergone hydrolysis.

(iii) This was well answered by most candidates. The majority of answers referred to esterification/substitution instead of condensation.

(iv) A small proportion of candidates answered this question correctly.

(v) A small proportion of candidates answered this question correctly.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**

3(d)(i)	increases the rate of reaction	**1**
	by providing an alternative reaction pathway of lower E_a	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.